Mapping Beauty-Supply Retail Access as an Exposure-Opportunity Proxy for Hair-Adhesive Products: A Census-Tract Geospatial Analysis of North Carolina

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Abstract

Hair adhesives (lace, wig, and eyelash bonding products) are marketed disproportionately to Black consumers and are applied in prolonged contact with the scalp, hairline, and eyelid, yet the geography of where these products are *obtained* has not been characterized. Because the retail point of purchase is an upstream, observable proxy for the opportunity to acquire and use a product, we mapped beauty-supply retail access across North Carolina and linked it to census-tract race and income. Store locations were extracted from OpenStreetMap, tract demographics from the U.S. Census American Community Survey (ACS) 2022 five-year estimates, and access was computed as distance to the nearest store, store count within a 5 km buffer, and a stores-per-capita access score, using an open-source GeoPandas pipeline with a parallel ArcGIS Pro (`arcpy`) workflow. Across 2,649 North Carolina tracts (population 10.5 million) served by 349 identified stores (183 inside the state), 67.5% of tracts had no beauty-supply store within 5 km (median distance to nearest store 9.3 km, IQR 3.9 to 20.4). Access was strongly spatially clustered (Moran's $I = 0.92$ for nearest-store distance and 0.83

for 5 km store count, both pseudo-$p = 0.001$). Contrary to a simple under-service hypothesis, access improved monotonically with tract percent-Black: tracts in the highest percent-Black quartile had a median nearest-store distance of 5.5 km versus 14.2 km in the lowest quartile, and 53.9% versus 81.7% lacked any store within 5 km. A negative-binomial model confirmed more stores in higher percent-Black tracts (incidence rate ratio 1.02 per percentage point, $p < 0.001$). We interpret this as spatial concentration of a targeted-market retail sector in Black and urban neighborhoods (greater *opportunity* for purchase and use) rather than equitable service, with large rural access gaps. These are ecological access measures and are not equivalent to purchase, use, dermal dose, or health outcome. The maps identify where hair-adhesive exposure research and risk communication would reach the most product-proximate populations.

Keywords: geospatial analysis; retail access; environmental justice; hair adhesives; beauty supply; spatial autocorrelation; GeoPandas; ArcGIS; North Carolina

1 Introduction

Hair adhesives (lace and wig bonding glues, bonding sprays, and eyelash adhesives) are liquid formulations engineered for sustained wear in direct contact with the scalp, hairline, eyelid, and adjacent facial skin. They are marketed heavily to, and used disproportionately by, Black women and girls, a population already shown to carry distinctive body burdens of endocrine-active and asthma-associated chemicals from hair-care products [Helm et al., 2018, James-Todd et al., 2021, Zota & Shamasunder, 2017]. Direct chemical measurement of consumer products, including non-targeted analysis of personal-care items, repeatedly recovers constituents that labels do not declare [Dodson et al., 2012, Boatman et al., 2026], so the exposure potential of these products is not fully described by their ingredient lists.

Hazard and exposure, however, both begin with *access*: a product must be obtainable before it can be purchased, applied, and absorbed. Where a product class is sold is therefore an upstream, observable determinant of exposure opportunity, and one that is unevenly distributed across neighborhoods. Retail access has been studied extensively for food [Walker et al., 2010] but rarely for targeted personal-care retail such as beauty-supply stores, the dominant point of purchase for hair adhesives. Geographic information systems (GIS) are the standard tool for quantifying such access and for testing whether it clusters by neighborhood race and income [Anselin, 1995, Moran, 1950].

This study maps beauty-supply retail access across North Carolina at the census tract level and asks three questions: (1) How far are North Carolina residents from a beauty-supply store, and how many such stores are near them? (2) Is that access spatially clustered? (3)

Does access vary systematically with tract percent-Black and poverty? We treat retail access strictly as an exposure-*opportunity* proxy, not as exposure itself, and we build the analysis in a fully reproducible open-source pipeline with a parallel ArcGIS Pro workflow.

2 Methods

2.1 Study area and unit of analysis

The primary analysis covers all census tracts in North Carolina with a nonzero population (2,649 of 2,672; 23 zero-population tracts were excluded). Five metropolitan case studies (Charlotte, Raleigh, Greensboro, Durham, and Fayetteville) are reported at finer grain as county subsets of the statewide layer. The unit of analysis is the census tract. All analyses are ecological: tract-level associations do not describe individuals, and access is not equated with purchase, use, internal dose, or disease.

2.2 Data sources

Store locations. Beauty-supply retail locations were extracted from OpenStreetMap via the Overpass API, filtered to shop categories (`cosmetics`, `perfumery`, `hairdresser_supply`) and to names matching a beauty-supply pattern (e.g., "Beauty Supply," "Sally," "Ulta," "Wig"), and narrowed to retail by excluding salon, spa, nail, and waxing service businesses. This yields a reproducible but incomplete census of stores (see Limitations).

Demographics. Tract demographics are U.S. Census ACS 2022 five-year estimates: total population and Black population (table B02001), median household income (B19013), and population for whom poverty status is determined and those below poverty (B17001). Tract geometries are 2022 TIGER/Line.

2.3 Access metrics

For each tract centroid we computed the Euclidean distance to the nearest store (BallTree nearest-neighbor query), the number of stores within a 5 km buffer, the percent of tract area within any store's 5 km service area, and a stores-per-capita access score,

$$\text{access score} = \frac{\text{stores within 5 km}}{\text{tract population}} \times 1000. \tag{1}$$

A composite underserved index used for intervention targeting was defined by the pipeline as

$$\text{underserved index} = 0.40\,\widetilde{B} + 0.25\,\widetilde{P} + 0.20\,\widetilde{A} + 0.15\,L, \tag{2}$$

where $\widetilde{B}$, $\widetilde{P}$ are min-max scaled percent-Black and poverty rate, $\widetilde{A}$ is a scaled low-access term combining inverse access score and distance, and L flags median income below \$40,000. Because this index embeds percent-Black with a 0.40 weight by construction, we do *not* use it as evidence of racial disparity; the disparity tests below use the raw access measures only, and the index is reported solely as a planning aid.

2.4 Spatial and statistical analysis

Global spatial autocorrelation of access was quantified with Moran's I under queen contiguity and row standardization, with inference by 999 random permutations [Moran, 1950, Anselin, 1995]. Associations between access and tract composition were summarized by Spearman rank correlation and by quartiles of percent-Black and of population density (persons per km^2). A negative-binomial generalized linear model (dispersion fixed at 1) regressed the 5 km store count on percent-Black, poverty rate, and median income (per \$10,000); it was re-estimated with maximum-likelihood dispersion and with county-clustered standard errors as sensitivity checks. Because Moran's I is tested by permutation, the smallest attainable pseudo-p with 999 permutations is 0.001. Underserved hotspots were delineated with DBSCAN on the high-need tracts.

2.5 Software and reproducibility

The open-source pipeline uses GeoPandas, scikit-learn (BallTree, DBSCAN), libpysal/esda (Moran's I), and statsmodels; a parallel ArcGIS Pro `arcpy` workflow in `scripts/` reproduces the proximity analysis in an Esri environment. (For clarity, "ArcGIS" denotes the Esri ArcGIS platform; it is not an acronym.) All code, input tables, and tract-level outputs are in the project repository, and the folder `paper_reproduction/` regenerates every number, table, and figure reported here (see Data and code availability).

3 Results

3.1 Statewide access

Across 2,649 North Carolina tracts (total population 10.5 million) served by 349 identified beauty-supply stores (183 inside the state; see Limitations), access was low and uneven:

67.5% of tracts had no store within 5 km, and the median distance to the nearest store was **9.3 km** (IQR 3.9 to 20.4 km). The statewide median tract was 14.7% Black. Figure 1 maps the distance surface with store locations: access concentrates in the Charlotte, Piedmont Triad, Triangle, and Fayetteville urban cores, with large far-from-store expanses across rural eastern and western counties.

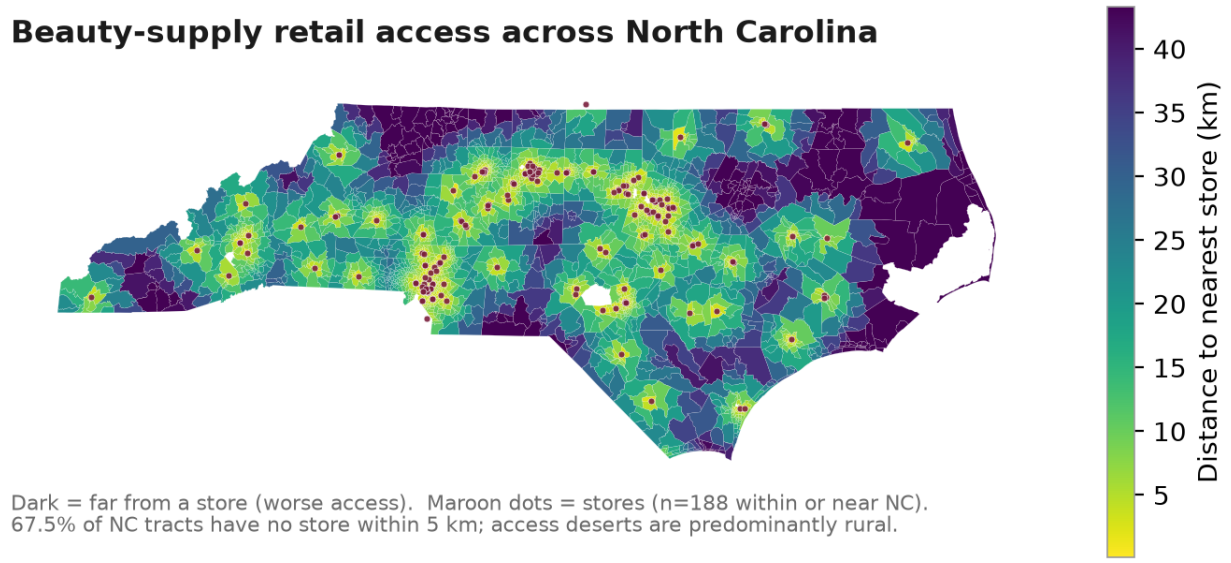


Figure 1: Distance from each census tract to the nearest beauty-supply store across North Carolina (ACS 2022 tracts; stores from OpenStreetMap). Darker tracts are farther from any store; maroon points are the 188 extracted stores inside the state or within about 8 km of it. Access clusters in urban cores while rural counties form contiguous access deserts.

3.2 Spatial clustering

Access was strongly and significantly clustered in space. Moran's I was **0.92** for nearest-store distance and **0.83** for 5 km store count (both pseudo-$p = 0.001$, 999 permutations), versus 0.66 for percent-Black. Retail access is thus even more spatially concentrated than the demographic pattern itself.

3.3 Access by neighborhood racial composition

Contrary to a simple under-service hypothesis, access *improved* as tract percent-Black increased (Table 1). Tracts in the highest percent-Black quartile were less than half as far from a store as those in the lowest quartile and were far more likely to have a store within 5 km.

Table 1: Beauty-supply access by quartile of tract percent-Black, North Carolina (2,649 tracts). Higher percent-Black tracts have closer and more numerous stores.

Percent-Black quartile	Tracts	Median nearest store (km)	Mean stores within 5 km	% tracts with no store in 5 km
Q1 (lowest % Black)	663	14.2	0.68	81.7
Q2	662	9.2	0.97	70.1
Q3	662	8.7	0.98	64.0
Q4 (highest % Black)	662	5.5	1.23	53.9

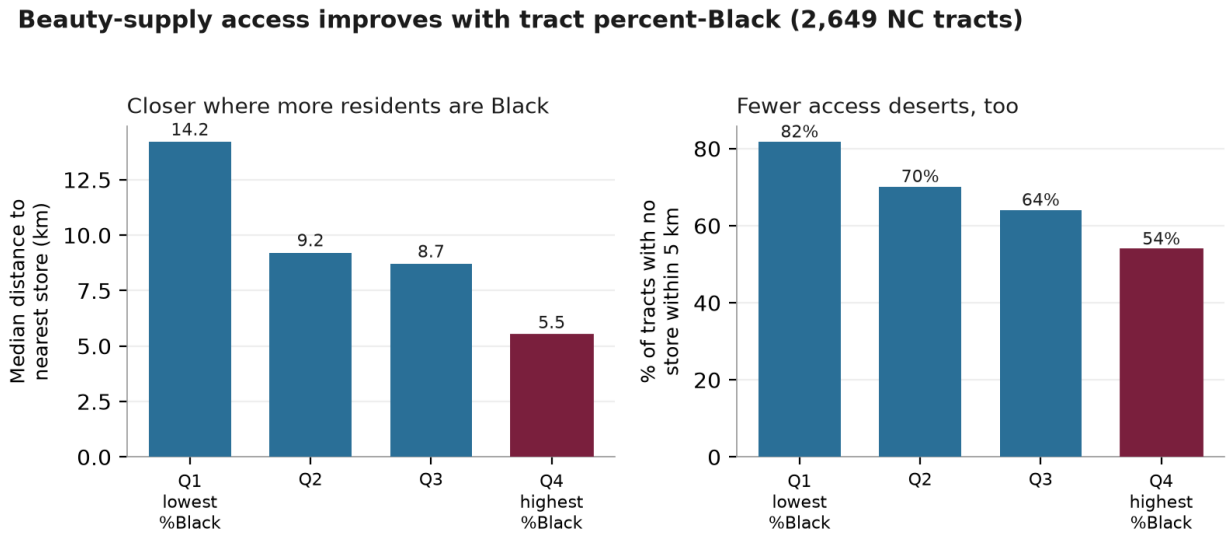


Figure 2: Beauty-supply access by quartile of tract percent-Black. Both distance to the nearest store (left) and the share of tracts with no store within 5 km (right) decrease monotonically as percent-Black rises; the highest-percent-Black quartile (maroon) has the best access.

Spearman correlations were consistent: percent-Black was negatively associated with distance to the nearest store ($\rho = -0.21$) and positively with store count ($\rho = 0.22$). In the negative-binomial model ($n = 2{,}624$ tracts with complete covariates; dispersion fixed at 1), the expected 5 km store count rose with percent-Black (incidence rate ratio [IRR] 1.02 per percentage point, $p < 0.001$), poverty rate (IRR 1.02, $p < 0.001$), and median income (IRR 1.21 per \$10,000, $p < 0.001$). The associations kept their direction and significance when dispersion was estimated by maximum likelihood (IRR 1.02 for percent-Black, 1.02 for poverty, 1.24 for income) and when standard errors were clustered by county (all $p < 0.001$). Access also tracked population density: 98.5% of tracts in the lowest-density quartile had no store within 5 km (median distance 22.6 km), compared with 24.2% (3.1 km) in the highest-density quartile. The percent-Black gradient therefore sits on top of a steep urban to rural gradient.

3.4 Metropolitan case studies

Five metropolitan areas were examined as county subsets of the statewide layer, so that the same store set and ACS vintage apply throughout (Table 2, Figure 3). Every metro had better access than the state as a whole: the share of tracts with no store within 5 km ranged from 22% in Charlotte to 54% in Fayetteville, against 67.5% statewide. Access remained significantly spatially clustered within every metro (Moran's I for nearest-store distance 0.61 to 0.79, all pseudo-$p = 0.001$). Within-metro racial gradients were modest and mixed. Majority-Black tracts were closer to the nearest store than other tracts in Greensboro (median 2.4 vs 3.4 km), Raleigh (3.4 vs 3.8 km), and Fayetteville (4.4 vs 5.7 km), and slightly farther in Charlotte (3.5 vs 3.1 km) and Durham (4.3 vs 3.5 km). This pattern suggests that the strong statewide gradient in Table 1 reflects, at least in part, the contrast between urban and rural counties rather than consistent sorting within metros.

Table 2: Metropolitan case studies (county subsets of the statewide layer, same 349-store extract and ACS 2022 vintage). "% no store" is the share of tracts with no store within 5 km. Moran's I is for nearest-store distance (pseudo-$p = 0.001$ in every row). Statewide shown for reference.

Area	Tracts	Median % Black	Median nearest (km)	% no store in 5 km	Mean stores ≤5 km	Moran's I
North Carolina	2,649	14.7	9.3	67.5	0.97	0.92
Charlotte	302	29.8	3.2	22.2	3.11	0.79
Raleigh	229	14.3	3.7	31.9	1.75	0.78
Greensboro	125	29.4	3.0	29.6	4.18	0.70
Durham	67	31.2	3.6	29.9	1.22	0.61
Fayetteville	81	35.9	5.2	54.3	1.33	0.76

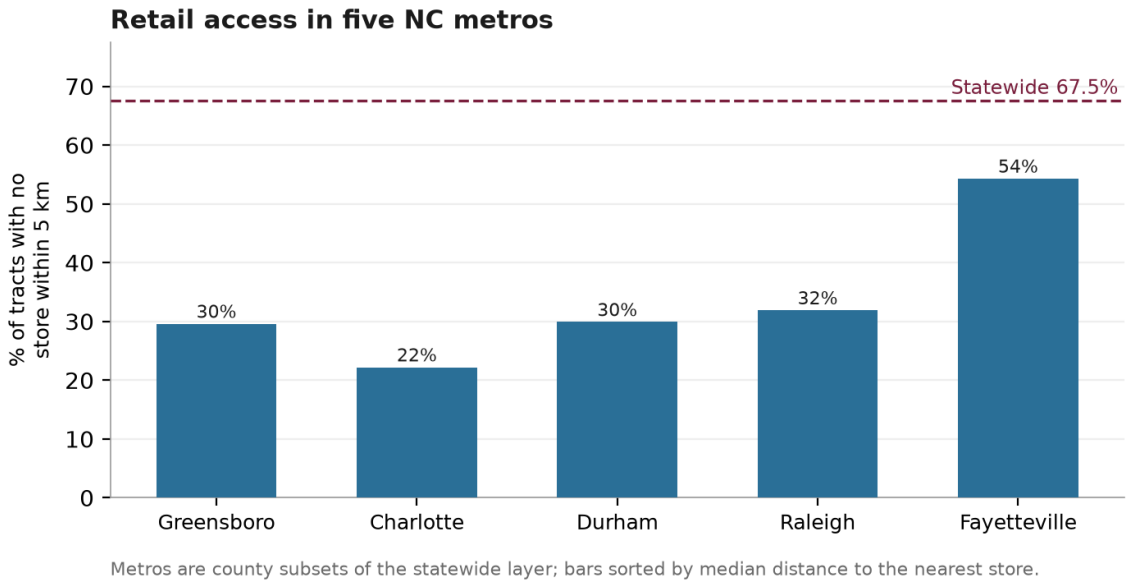


Figure 3: Share of tracts with no beauty-supply store within 5 km across five North Carolina metros (county subsets of the statewide layer). Every metro has better access than the state as a whole (dashed line, 67.5%).

4 Discussion

The clearest finding is that beauty-supply retail in North Carolina is highly spatially concentrated and is *more* available in Black and urban neighborhoods, while two-thirds of the state's tracts, disproportionately rural, lack a store within 5 km. This is consistent with beauty-supply stores functioning as a targeted-market retail sector that locates where demand for products such as hair adhesives is greatest. Framed as exposure opportunity, the geography means the point-of-purchase for these prolonged-contact products is most dense precisely in the communities already identified in the biomonitoring literature as carrying elevated hair-product chemical burdens [Helm et al., 2018, James-Todd et al., 2021, Zota & Shamasunder, 2017]. Greater availability is not the same as greater risk, but it is where the opportunity for acquisition and repeated use is concentrated, and therefore where exposure research and risk communication would reach the most product-proximate populations.

This reframing matters because the naive environmental-justice expectation that Black neighborhoods are *under*-served does not hold for this retail class; the data contradict it. Reporting the observed direction is itself part of the contribution: access disparities here follow an urban to rural contrast layered on a targeted-market concentration, not a simple race-based deficit.

4.1 Limitations

1. **Store data are incomplete and include out-of-state stores.** Locations come from OpenStreetMap, a volunteer dataset that undercounts stores, especially in rural areas and for independent retailers, and store counts varied across extraction runs. The statewide extract appears to have been retrieved with a bounding-box query: 161 of its 349 points lie more than about 8 km outside North Carolina, and 156 of those fall inside the state's bounding box (for example stores in Greenville and Spartanburg, South Carolina), leaving 183 stores inside the state. Distances were computed against all 349 points, which lets tracts near the state line use the nearest store across it, but the figure of 349 overstates the number of North Carolina stores. Only stores inside or within about 8 km of the state are drawn in Figure 1. The absolute access figures are therefore approximate.

2. **Ecological design.** Tract-level access does not describe individuals and is not purchase, use, dermal dose, or disease. No causal or individual inference is warranted.

3. **Access is a crude proxy.** Distance from a tract centroid to the nearest store, computed on planar coordinates in a single projected frame, ignores road networks, travel mode, online purchasing, and within-tract heterogeneity, and introduces mild geometric distortion over the full state extent.

4. **Composite index is race-weighted by construction.** The underserved index embeds percent-Black at a 0.40 weight and is used only for intervention targeting, never as evidence of disparity; all disparity results rest on raw access measures.

5. **Cross-sectional.** A single ACS vintage and a single store snapshot cannot speak to change over time.

6. **No product verification.** Store categories were not field-verified to stock hair adhesives specifically.

4.2 Implications and future work

The maps provide a site-selection tool: the highest-access, higher-percent-Black urban clusters are where recruitment for product sampling, biomonitoring, or salon-worker studies would be most efficient, and where plain-language risk communication would reach the most users. Priority extensions are (1) replacing OpenStreetMap counts with a validated store registry; (2) network- rather than straight-line distance; and (3) linking this access layer to

the product-level chemical and dermal-exposure analysis developed elsewhere in this project, so that *where* products are obtained can be joined to *what* is in them.

5 Conclusion

Beauty-supply retail access in North Carolina is strongly spatially clustered and concentrated in Black and urban neighborhoods, with large rural access gaps. Interpreted as an exposure-opportunity proxy for hair-adhesive products, the geography locates the communities most product-proximate to a class of prolonged-contact cosmetics. This offers an evidence-based foundation for targeting exposure research and risk communication, reported in the direction the data show rather than the one assumed.

Data and code availability

All code, input tables, and tract-level outputs are available in the project repository. The folder `paper_reproduction/` regenerates every statistic, table, and figure in this paper from those files (`python reproduce.py`), and `verify_against_paper.py` checks the numbers printed here against the computed results.

Acknowledgments

Store data © OpenStreetMap contributors (ODbL); demographic data from the U.S. Census Bureau American Community Survey.

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